

```

1 //@version=5
2 // =====
3 // AURUM EDGE V4 - XAUUSD 30m - Symmetric Dual-Engine
4 //
5 // V3's long engine, untouched, plus two short engines built as structural
6 // mirrors of it. All three are independently switchable.
7 //
8 // -----
9 // READ THIS BEFORE YOU ENABLE THE SHORT ENGINES
10 //
11 // Both short engines are OFF by default, and that is a measurement, not caution.
12 // They were built to spec and tested on 1,678,872 one-minute bars (Jan 2021 -
13 // Aug 2026) with real historical spreads, and the daily bear gate was ACTIVE FOR
14 // 18.6% of that history (10,411 of 55,988 30m bars, longest run 123 days). So
15 // this was a fair test on a real sample, not an untested regime.
16 //
17 // SYMMETRIC SHORT ENGINE (daily bear gate, 3 mirrored triggers)
18 // mirrored ladder exit n=491 E[R] -0.086 PF 0.85 t=-1.28
19 // BE@1R + 2R target n=638 E[R] -0.007 PF 0.99 t=-0.14 <- least bad
20 // fixed 1R target n=736 E[R] -0.024 PF 0.95 t=-0.65
21 // 4xATR trail n=573 E[R] -0.080 PF 0.84 t=-1.56
22 // on 1h bars n=321 E[R] -0.136 PF 0.78
23 // on 4h bars n=117 E[R] -0.215 PF 0.68
24 //
25 // COUNTER-TREND SHORT (overbought spike inside a bull regime)
26 // ALL 18 variants tested were negative, most significantly so:
27 // ext>1.5ATR RSI>70 tp1R n=575 E[R] -0.206 t=-5.19 p=1.000
28 // ext>2.0ATR RSI>75 ladder n=225 E[R] -0.399 t=-5.31 p=1.000
29 //
30 // WHY. Gold is not symmetric. It carries structural upward drift, so its
31 // downtrends are shallower, choppiest and revert faster than its uptrends. A
32 // mirrored trend-follower assumes downmoves persist the way upmoves do; on gold
33 // they do not. The pullbacks that look like short setups on a daily chart are
34 // mostly mean-reverting dips inside an uptrend - which is precisely what the
35 // LONG engine is buying, and why shorting them loses.
36 //
37 // The engines are here, complete and symmetric, so you can forward-test them
38 // yourself and so the machinery exists if gold ever breaks its multi-year
39 // uptrend. At that point the short side is worth re-testing on new data. Until
40 // then, the evidence says leave them off.
41 //
42 // -----
43 // LONG ENGINE - walk-forward validated, out-of-sample only
44 // 16 folds, Jan 2023 - Mar 2026, exit re-chosen inside each training window
45 // 2.40 trades/day n=2011 E[R] +0.284 R total +572.0 R
46 // win 34.2% PF 1.50 maxDD 47.5 R t=+6.83 p<0.0001
47 // deflated Sharpe P(SR>0) = 1.000
48 // At 0.25% risk: ~+48%/yr at ~13% drawdown, against buy-and-hold gold at
49 // +47%/yr and 19% drawdown over the same window.
50 //
51 // Caveat, stated plainly: all 2,011 trades are long and the out-of-sample
52 // window is a gold bull market. The least-trending slice (2023) returned
53 // +0.118 R at p=0.066, which is not significant.
54 // =====
55
56 strategy(
57 "Aurum Edge V4 - XAUUSD 30m Dual Engine",
58 overlay = true,
59 pyramiding = 10,
60 initial_capital = 10000,
61 currency = currency.USD,
62 default_qty_type = strategy.percent_of_equity,
63 default_qty_value = 100,
64 process_orders_on_close = false,
65 calc_on_every_tick = false,
66 calc_on_order_fills = false,
67 commission_type = strategy.commission.cash_per_contract,
68 commission_value = 0.02,
69 slippage = 3,
70 max_labels_count = 500)
71
72 // =====
73 // INPUTS
74 // =====
75 string G_ENG = "1. Engines"
76 string G_R = "2. Risk"
77 string G_LE = "3. Long entries (bull regime)"
78 string G_SE = "4. Short entries (bear regime)"
79 string G_CT = "5. Counter-trend shorts (bull overbought)"
80 string G_LAD = "6. Ladder exit"
81 string G_UI = "7. Display and alerts"
82
83 enableLong = input.bool(true, "Enable Long Engine (bull regime)", group = G_ENG)
84 enableShort = input.bool(false, "Enable Symmetric Short Engine (bear regime)", group = G_ENG,
85 tooltip = "OFF by default because it was measured at -0.086 R per trade over 491 " +
86 "trades in a genuine bear sample (18.6% of history). Enable to forward-test.")
87 enableCT = input.bool(false, "Enable Counter-Trend Short Engine (bull overbought)", group = G_ENG,
88 tooltip = "OFF by default. All 18 tested variants were negative, most at t < -4. " +
89 "Shorting overbought gold inside an uptrend lost money reliably.")

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90
91 riskPct    = input.float(0.25, "Risk per trade (% of equity)", 0.05, 3.0, 0.05, group = G_R)
92 ctRiskMult = input.float(0.5, "Counter-trend risk multiplier", 0.1, 1.0, 0.1, group = G_R,
93     tooltip = "Counter-trend trades size smaller than trend trades.")
94 stopAtr    = input.float(2.0, "Initial stop (ATR)", 0.5, 6.0, 0.1, group = G_R)
95 minStopUsd = input.float(8.0, "Minimum stop ($)", 1.0, 60.0, 0.5, group = G_R,
96     tooltip = "Cost drag = round-trip cost / stop distance. Round trip on gold is " +
97         "roughly $0.45-$1.20; below about $5 of stop the spread eats the edge.")
98 maxStopUsd = input.float(200.0, "Maximum stop ($)", 20.0, 500.0, 5.0, group = G_R)
99 maxOpen    = input.int(5, "Max concurrent positions (per side)", 1, 10, group = G_R)
100
101 useL1 = input.bool(true, "L1: pullback to EMA20", group = G_LE)
102 useL2 = input.bool(true, "L2: deeper pullback to EMA50", group = G_LE)
103 useL3 = input.bool(true, "L3: 24-bar breakout", group = G_LE)
104
105 useS1 = input.bool(true, "S1: rally pullback to EMA20", group = G_SE)
106 useS2 = input.bool(true, "S2: deep rally pullback to EMA50", group = G_SE)
107 useS3 = input.bool(true, "S3: 24-bar breakdown", group = G_SE)
108 shortExitMode = input.string("Tight 1R target", "Short exit style", group = G_SE,
109     options = ["Mirrored ladder", "Tight 1R target", "BE at 1R then 2R target"],
110     tooltip = "Trend-following exits were the WORST on shorts (-0.086 R) while a tight " +
111         "target was least bad (-0.007 R). The asymmetry is the point: gold's " +
112         "downmoves do not persist the way its upmoves do.")
113
114 ctExtAtr = input.float(2.0, "Extension above EMA20 (ATR)", 0.5, 6.0, 0.25, group = G_CT)
115 ctRsi    = input.float(70, "RSI threshold", 50, 95, 1, group = G_CT)
116 ctTpR    = input.float(1.0, "Counter-trend target (R)", 0.5, 5.0, 0.25, group = G_CT)
117
118 r1Trig = input.float(1.5, "Rung 1 trigger (R)", 0.25, 5.0, 0.25, group = G_LAD)
119 r1Lock = input.float(0.0, "Rung 1 lock (R)", -1.0, 5.0, 0.1, group = G_LAD, tooltip = "0 = breakeven")
120 r2Trig = input.float(3.0, "Rung 2 trigger (R)", 0.5, 9.0, 0.25, group = G_LAD)
121 r2Lock = input.float(1.5, "Rung 2 lock (R)", -1.0, 9.0, 0.1, group = G_LAD)
122 r3Trig = input.float(5.0, "Rung 3 trigger (R)", 1.0, 15.0, 0.25, group = G_LAD)
123 r3Lock = input.float(3.2, "Rung 3 lock (R)", -1.0, 15.0, 0.1, group = G_LAD)
124 maxHoldBars = input.int(96, "Time stop (bars)", 8, 500, group = G_LAD, tooltip = "96 x 30m = 48h")
125
126 atrLen    = input.int(14, "ATR length", 2, 100, group = G_UI)
127 showLabels = input.bool(true, "Show trade labels", group = G_UI)
128 showPanel = input.bool(true, "Show regime panel", group = G_UI)
129 tintBg    = input.bool(true, "Tint background by regime", group = G_UI)
130
131 // =====
132 // HIGHER-TIMEFRAME REGIME GATES
133 // Only COMPLETED daily bars are read ([1] offset), so these cannot repaint.
134 // =====
135 dE20 = request.security(syminfo.tickerid, "D", ta.ema(close, 20)[1], lookahead = barmerge.lookahead_on)
136 dE50 = request.security(syminfo.tickerid, "D", ta.ema(close, 50)[1], lookahead = barmerge.lookahead_on)
137 dE200 = request.security(syminfo.tickerid, "D", ta.ema(close, 200)[1], lookahead = barmerge.lookahead_on)
138
139 bool macroBull = dE20 > dE50 and dE50 > dE200 // long gate
140 bool macroBear = dE20 < dE50 and dE50 < dE200 // exact inverse
141
142 // =====
143 // 30-MINUTE ENGINE
144 // =====
145 ema20 = ta.ema(close, 20)
146 ema50 = ta.ema(close, 50)
147 ema200 = ta.ema(close, 200)
148 atr    = ta.atr(atrLen)
149 rsi    = ta.rsi(close, 14)
150
151 plot(ema20, "EMA20", color = color.new(color.aqua, 0), linewidth = 1)
152 plot(ema50, "EMA50", color = color.new(color.orange, 0), linewidth = 1)
153 plot(ema200, "EMA200", color = color.new(color.purple, 0), linewidth = 2)
154
155 bool stackUp = ema20 > ema50 and ema50 > ema200
156 bool stackDn = ema20 < ema50 and ema50 < ema200
157 bool slopeUp = ema20 > ema20[3]
158 bool slopeDn = ema20 < ema20[3]
159
160 float hh24 = ta.highest(high, 24)[1]
161 float ll24 = ta.lowest(low, 24)[1]
162
163 // ---- LONG triggers (identical to V3) ----
164 bool sigL1 = useL1 and stackUp and slopeUp and low <= ema20 and close > ema20 and close > open
165 bool sigL2 = useL2 and stackUp and low <= ema50 and close > ema50 and close > open
166 bool sigL3 = useL3 and close > hh24 and close > open
167 bool longSignal = enableLong and macroBull and (sigL1 or sigL2 or sigL3)
168
169 // ---- SHORT triggers (exact structural mirrors) ----
170 bool sigS1 = useS1 and stackDn and slopeDn and high >= ema20 and close < ema20 and close < open
171 bool sigS2 = useS2 and stackDn and high >= ema50 and close < ema50 and close < open
172 bool sigS3 = useS3 and close < ll24 and close < open
173 bool shortSignal = enableShort and macroBear and (sigS1 or sigS2 or sigS3)
174
175 // ---- COUNTER-TREND short (overbought exhaustion inside a bull regime) ----
176 float extAtr = atr > 0 ? (close - ema20) / atr : 0.0
177 bool sigCT = enableCT and macroBull and extAtr >= ctExtAtr and rsi >= ctRsi and close < open
178

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179 // =====
180 // POSITION SIZING
181 // =====
182 float stopDist = math.min(math.max(atr * stopAtr, minStopUsd), maxStopUsd)
183 float qtyBase = stopDist > 0 ? (strategy.equity * riskPct / 100.0) / stopDist : na
184 float qtyCT = na(qtyBase) ? na : qtyBase * ctRiskMult
185
186 int openLongs = 0
187 int openShorts = 0
188 if strategy.opentrades > 0
189     for i = 0 to strategy.opentrades - 1
190         if strategy.opentrades.size(i) > 0
191             openLongs += 1
192         else
193             openShorts += 1
194
195 // =====
196 // ENTRIES - each position carries its own id so it can run its own ladder
197 // =====
198 var int seq = 0
199 var map<string, float> riskOf = map.new<string, float>()
200 var map<string, float> entryOf = map.new<string, float>()
201 var map<string, int> barOf = map.new<string, int>()
202 var map<string, int> kindOf = map.new<string, int>() // 0 long, 1 short, 2 counter-trend
203
204 _register(string id, float rp, float ep, int kind) =>
205     map.put(riskOf, id, rp)
206     map.put(entryOf, id, ep)
207     map.put(barOf, id, bar_index)
208     map.put(kindOf, id, kind)
209
210 if longSignal and openLongs < maxOpen and not na(qtyBase) and qtyBase > 0
211     seq += 1
212     string id = "L" + str.toString(seq)
213     strategy.entry(id, strategy.long, qty = qtyBase)
214     _register(id, stopDist, close, 0)
215     if showLabels
216         label.new(bar_index, low, "L", style = label.style_label_up,
217             color = color.new(color.green, 20), textcolor = color.white, size = size.tiny)
218     alert("AURUM_V4 LONG entry=" + str.toString(close, "#.##") +
219         " stop=" + str.toString(close - stopDist, "#.##") +
220         " risk=" + str.toString(stopDist, "#.##"), alert.freq_once_per_bar_close)
221
222 if shortSignal and openShorts < maxOpen and not na(qtyBase) and qtyBase > 0
223     seq += 1
224     string id = "S" + str.toString(seq)
225     strategy.entry(id, strategy.short, qty = qtyBase)
226     _register(id, stopDist, close, 1)
227     if showLabels
228         label.new(bar_index, high, "S", style = label.style_label_down,
229             color = color.new(color.red, 20), textcolor = color.white, size = size.tiny)
230     alert("AURUM_V4 SHORT entry=" + str.toString(close, "#.##") +
231         " stop=" + str.toString(close + stopDist, "#.##") +
232         " risk=" + str.toString(stopDist, "#.##"), alert.freq_once_per_bar_close)
233
234 if sigCT and openShorts < maxOpen and not na(qtyCT) and qtyCT > 0
235     seq += 1
236     string id = "C" + str.toString(seq)
237     strategy.entry(id, strategy.short, qty = qtyCT)
238     _register(id, stopDist, close, 2)
239     if showLabels
240         label.new(bar_index, high, "CT", style = label.style_label_down,
241             color = color.new(color.orange, 20), textcolor = color.white, size = size.tiny)
242     alert("AURUM_V4 CT-SHORT entry=" + str.toString(close, "#.##") +
243         " stop=" + str.toString(close + stopDist, "#.##"), alert.freq_once_per_bar_close)
244
245 // =====
246 // EXITS - per position, ratcheting on CLOSED BARS ONLY
247 //
248 // The ratchet reads the completed bar's close, never its running high/low. That
249 // is deliberate and is why calc_on_every_tick is false. Trailing from an
250 // intrabar extreme lets the stop jump to a level the trade never actually held;
251 // in testing that produced a +0.089 R phantom edge on coin-flip entries with
252 // costs switched off, where the true answer is mathematically zero.
253 // =====
254 if strategy.opentrades > 0
255     for i = 0 to strategy.opentrades - 1
256         string id = strategy.opentrades.entry_id(i)
257         if map.contains(riskOf, id)
258             float rp = map.get(riskOf, id)
259             float ep = map.get(entryOf, id)
260             int eb = map.get(barOf, id)
261             int kind = map.get(kindOf, id)
262             int dir = kind == 0 ? 1 : -1
263
264             if rp > 0
265                 float excursionR = (close - ep) * dir / rp
266                 float stopLvl = ep - rp * dir // initial stop
267                 float tgtLvl = na

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268
269         if kind == 0 // LONG: full ladder
270             if excursionR >= r3Trig
271                 stopLvl := ep + r3Lock * rp
272             else if excursionR >= r2Trig
273                 stopLvl := ep + r2Lock * rp
274             else if excursionR >= r1Trig
275                 stopLvl := ep + r1Lock * rp
276
277         else if kind == 1 // SHORT: chosen style
278             if shortExitMode == "Mirrored ladder"
279                 if excursionR >= r3Trig
280                     stopLvl := ep - r3Lock * rp
281                 else if excursionR >= r2Trig
282                     stopLvl := ep - r2Lock * rp
283                 else if excursionR >= r1Trig
284                     stopLvl := ep - r1Lock * rp
285             else if shortExitMode == "Tight 1R target"
286                 tgtLvl := ep - rp
287             else // BE at 1R then 2R target
288                 tgtLvl := ep - 2.0 * rp
289                 if excursionR >= 1.0
290                     stopLvl := ep
291
292         else // COUNTER-TREND short
293             tgtLvl := ep - ctTpR * rp
294
295         if na(tgtLvl)
296             strategy.exit("x" + id, from_entry = id, stop = stopLvl)
297         else
298             strategy.exit("x" + id, from_entry = id, stop = stopLvl, limit = tgtLvl)
299
300         if bar_index - eb >= maxHoldBars
301             strategy.close(id, comment = "time stop")
302
303 // =====
304 // DISPLAY
305 // =====
306 bgcolor(tintBg and macroBull ? color.new(color.green, 94) :
307         tintBg and macroBear ? color.new(color.red, 94) : na, title = "Regime")
308
309 plotshape(longSignal, "Long signal", shape.triangleup, location.belowbar,
310         color.new(color.green, 0), size = size.tiny)
311 plotshape(shortSignal, "Short signal", shape.triangledown, location.abovebar,
312         color.new(color.red, 0), size = size.tiny)
313 plotshape(sigCT, "CT signal", shape.xcross, location.abovebar,
314         color.new(color.orange, 0), size = size.tiny)
315
316 var table t = table.new(position.top_right, 2, 8, border_width = 1)
317 if barstate.islast and showPanel
318     table.cell(t, 0, 0, "Aurum Edge V4", text_color = color.white,
319         bgcolor = color.new(color.teal, 25))
320     table.cell(t, 1, 0, "", bgcolor = color.new(color.teal, 25))
321     table.cell(t, 0, 1, "Daily regime")
322     table.cell(t, 1, 1, macroBull ? "BULL" : macroBear ? "BEAR" : "neutral",
323         text_color = macroBull ? color.green : macroBear ? color.red : color.gray)
324     table.cell(t, 0, 2, "Long engine")
325     table.cell(t, 1, 2, enableLong ? (macroBull ? "armed" : "idle") : "off")
326     table.cell(t, 0, 3, "Short engine")
327     table.cell(t, 1, 3, enableShort ? (macroBear ? "armed" : "idle") : "off (tested -0.086R)")
328     table.cell(t, 0, 4, "Counter-trend")
329     table.cell(t, 1, 4, enableCT ? "armed" : "off (tested -0.21R)")
330     table.cell(t, 0, 5, "Open L / S")
331     table.cell(t, 1, 5, str.tostring(openLongs) + " / " + str.tostring(openShorts))
332     table.cell(t, 0, 6, "Stop ($) / cost%")
333     table.cell(t, 1, 6, str.tostring(stopDist, "#.##") + " / " +
334         str.tostring(0.45 / stopDist * 100, "#.##") + "%")
335     table.cell(t, 0, 7, "Closed trades")
336     table.cell(t, 1, 7, str.tostring(strategy.closedtrades))
337
338 // =====
339 // ALERT CONDITIONS (for automation wiring)
340 // =====
341 alertcondition(longSignal, "Aurum V4 Long entry", "Aurum Edge V4: LONG {{ticker}} @ {{close}}")
342 alertcondition(shortSignal, "Aurum V4 Short entry", "Aurum Edge V4: SHORT {{ticker}} @ {{close}}")
343 alertcondition(sigCT, "Aurum V4 CT short", "Aurum Edge V4: COUNTER-TREND SHORT {{ticker}} @ {{close}}")

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